DataLab: Grow Your Own Psychology Data

PS50008A Week 11 Lab Activity

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16 January 2026

Overview

! Welcome to DataLab!

Today you'll complete a memory test together, then work through 5 tasks using your sequence card. You'll generate the dataset we analyse throughout the term.

Session Structure (2 hours)

Time	Activity
10:00–10:25	Welcome, Part 1 survey, **Memory Test** (together)
10:25–11:45	Complete 5 tasks (follow your sequence card)
11:45–12:00	Part 2 survey & wrap-up

Before You Start

1. **Collect your participant card** from the instructors
2. **Collect your sequence card** — this tells you which tasks to do in which order
3. **Find a partner** with the same colour
4. **Scan the QR code** and complete Part 1 of the Qualtrics survey
5. **STOP** when you reach the Memory Test page — we'll do that together!

Part 1 Survey

Before the memory test, you'll complete:

- Consent form
- Basic information (sleep, lifestyle)
- How you're feeling right now (mood, energy, stress)
- A brief personality questionnaire (TIPI)
- Demographics

Memory Test: Mind Palace

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MIND PALACE

Memory test materials

STEP 1: DISTRACTOR TASK (DO THIS FIRST, FOR 60 SECONDS)

Complete as many as you can. This prevents rehearsal of the word list.

47 + 36 =	83 - 29 =	6 × 7 =	54 ÷ 9 =	28 + 45 =
54 + 18 =	91 - 47 =	8 × 9 =	72 ÷ 8 =	63 + 19 =
26 + 57 =	72 - 38 =	4 × 12 =	45 ÷ 5 =	37 + 26 =
63 + 29 =	85 - 56 =	7 × 8 =	63 ÷ 7 =	52 + 31 =
38 + 45 =	64 - 27 =	9 × 6 =	48 ÷ 6 =	44 + 29 =
52 + 39 =	93 - 48 =	5 × 11 =	81 ÷ 9 =	67 + 18 =

STEP 2: WORD RECALL (WRITE ALL THE WORDS YOU REMEMBER)

1	2	3	4	5
6	7	8	9	10

Write words in any order — the numbers are just to help you count.

✍ **TOTAL WORDS RECALLED: _____ / 10**

What survives the interference?

We complete the memory test **together** as a group. Two word lists will appear on screen — **List A on the LEFT** and **List B on the RIGHT**. Your instructor will tell you which side to look at.

The Procedure

1. **STUDY (30 sec)**: Look at YOUR word list and memorise as many as you can
2. **DISTRACT (60 sec)**: Complete maths problems (on screen)
3. **RECALL (90 sec)**: Write down as many words as you remember on paper

4. **RECORD**: Enter your results in Qualtrics

What You'll Record

- Total words correctly recalled (out of 10)
- Which positions (1–10) you remembered
- Any "intrusions" (words you wrote that weren't on the list)
- What strategy (if any) you used

> ⚠ Important
>
> Only look at YOUR assigned list (left or right). Write your answers on paper during recall — don't look back at the screen!

> Data Dictionary
>
> See the Data Dictionary for a complete reference of all variables you'll be collecting.

The Five Tasks

After the memory test, follow your **sequence card** to complete these 5 tasks (~15 minutes each):

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REFLEX TEST

How fast is your nervous system?

INSTRUCTIONS

1. **Partner A** holds ruler at 30cm mark, hanging vertically
2. **Partner B** positions thumb and finger at 0cm — don't touch the ruler!
3. A drops ruler **without warning**
4. B catches it **as fast as possible**
5. Read the cm mark at the **top of your thumb**
6. Complete **5 trials**, then swap roles

CONVERSION CHART (CM → MILLISECONDS)

cm	1	2	3	4	5	6	7	8	9	10
ms	45	64	78	90	101	111	120	128	135	143
cm	11	12	13	14	15	16	17	18	19	20
ms	150	156	163	169	175	181	186	192	197	202
cm	21	22	23	24	25	26	27	28	29	30
ms	207	212	217	221	226	230	235	239	243	247

Formula: RT (ms) = $\sqrt{(2d \div 9.81)} \times 1000$ where d = distance in metres

✍ RECORD YOUR CATCHES (CM)

Trial 1	Trial 2	Trial 3	Trial 4	Trial 5

Which hand did you use? ☐ Dominant ☐ Non-dominant

Figure 1: Task 1: Reflex Test

DATALAB

TIME WARP

Your brain's internal clock

TASK A: TIME ESTIMATION

1. Close your eyes or look down at the desk
2. When you hear **"Start"**, begin sensing time passing
3. Say **"Stop"** when you believe exactly **60 seconds** have passed
4. Your partner will record your actual time

TASK B: LINE LENGTH

1. Look at **Line A** (separate sheet)
2. Estimate length in cm
3. Repeat for **Line B**

Don't measure!

TASK C: STRING LENGTH

1. Feel **String 1** (loop) without looking
2. Estimate length in cm
3. Repeat for **String 2** (knotted)

Touch only!

TASK D: TEMPERATURE

1. Without checking any devices
2. Estimate room temperature
3. Record in °C

✍ **RECORD YOUR ESTIMATES**

Time:	Line A:	Line B:	String 1:	String 2:	Temp:
_____ **sec**	_____ **cm**	_____ **cm**	_____ **cm**	_____ **cm**	_____ **°C**

Figure 2: Task 2: Time Warp

DATALAB

SIXTH SENSE

Can you feel when someone is watching?

INSTRUCTIONS

1. **TARGET** sits with back to partner, eyes closed
2. **STARER** flips a coin: **Heads** = stare at back of partner's head; **Tails** = look away
3. Hold for **15 seconds**, then tap partner's shoulder
4. Target says **"Staring"** or **"Not staring"** and rates confidence
5. Starer reveals the answer; record if correct
6. Complete **6 trials** as target, then swap roles

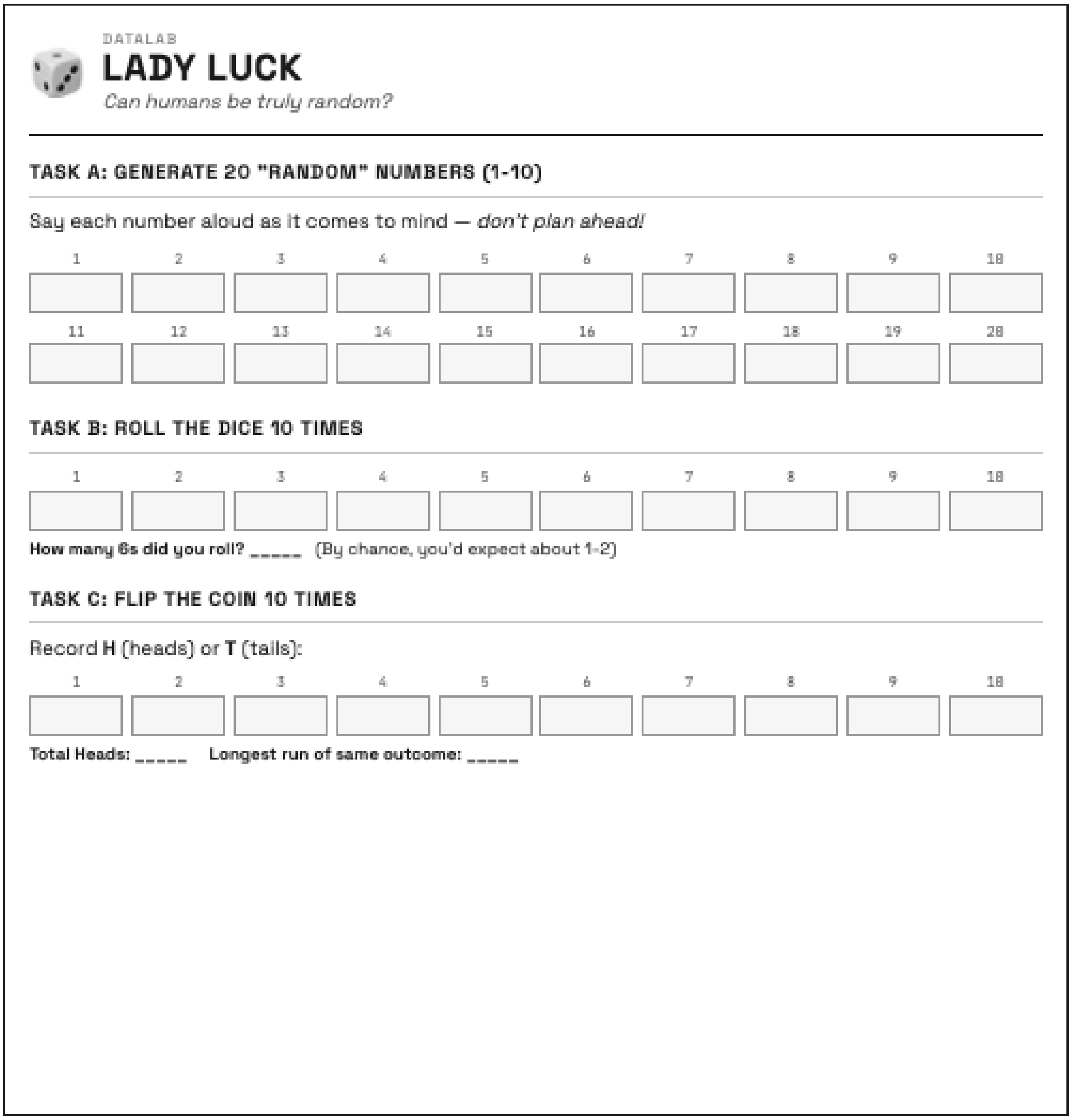
RECORDING SHEET

Trial	Actual	Your Guess	Confidence	Correct?
1	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
2	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
3	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
4	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
5	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕
6	☐ Staring ☐ Not	☐ Staring ☐ Not	1 2 3 4 5	☐ ✓ ☐ ✕

Confidence: 1 = pure guess, 5 = absolutely certain

✍ **TOTAL CORRECT: _____ / 6**

Figure 3: Task 3: Sixth Sense



DATALAB

LADY LUCK

Can humans be truly random?

TASK A: GENERATE 20 "RANDOM" NUMBERS (1-10)

Say each number aloud as it comes to mind — *don't plan ahead!*

1	2	3	4	5	6	7	8	9	10

11	12	13	14	15	16	17	18	19	20

TASK B: ROLL THE DICE 10 TIMES

1	2	3	4	5	6	7	8	9	10

How many 6s did you roll? _____ (By chance, you'd expect about 1-2)

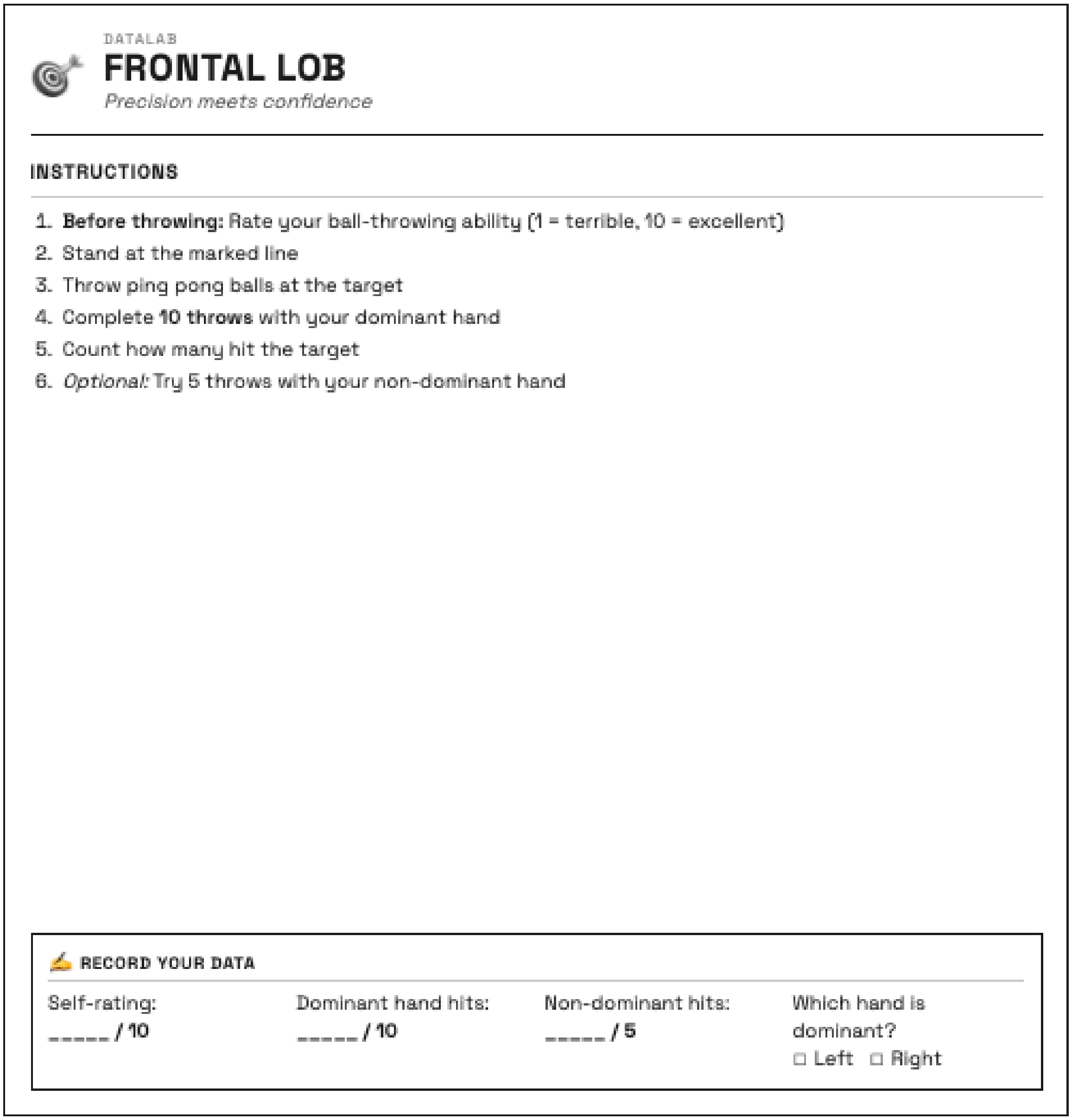
TASK C: FLIP THE COIN 10 TIMES

Record **H** (heads) or **T** (tails):

1	2	3	4	5	6	7	8	9	10

Total Heads: _____ **Longest run of same outcome:** _____

Figure 4: Task 4: Lady Luck



DATALAB

FRONTAL LOB

Precision meets confidence

INSTRUCTIONS

1. **Before throwing:** Rate your ball-throwing ability (1 = terrible, 10 = excellent)
2. Stand at the marked line
3. Throw ping pong balls at the target
4. Complete **10 throws** with your dominant hand
5. Count how many hit the target
6. *Optional:* Try 5 throws with your non-dominant hand

✍️ RECORD YOUR DATA

Self-rating:	Dominant hand hits:	Non-dominant hits:	Which hand is dominant?
______ **/ 10**	______ **/ 10**	______ **/ 5**	☐ Left ☐ Right

Figure 5: Task 5: Frontal Lob

Task 1: Reflex Test

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REFLEX TEST

How fast is your nervous system?

INSTRUCTIONS

1. **Partner A** holds ruler at 30cm mark, hanging vertically
2. **Partner B** positions thumb and finger at 0cm — don't touch the ruler!
3. A drops ruler **without warning**
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Formula: $RT\ (ms) = \sqrt{(2d \div 9.81)} \times 1000$ where d = distance in metres

✍ **RECORD YOUR CATCHES (CM)**

Trial 1	Trial 2	Trial 3	Trial 4	Trial 5

Which hand did you use? ☐ Dominant ☐ Non-dominant

How fast is your nervous system?

Measure your **reaction time** using a ruler-drop task.

What You'll Do

1. **Partner A** holds the ruler at the 30cm mark, letting it hang vertically
2. **Partner B** positions thumb and finger at the 0cm mark — don't touch!
3. Partner A drops the ruler **without warning** (within 5 seconds)
4. Partner B catches it **as fast as possible**
5. Read the cm mark at the **top of your thumb**

6. Complete **5 trials**, then swap roles

What You'll Record

- Which hand you used (dominant or non-dominant)
- The distance caught on each trial (in cm)
- Any trials you missed

Task 2: Time Warp

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TIME WARP

Your brain's internal clock

TASK A: TIME ESTIMATION

1. Close your eyes or look down at the desk
2. When you hear **"Start"**, begin sensing time passing
3. Say **"Stop"** when you believe exactly **60 seconds** have passed
4. Your partner will record your actual time

TASK B: LINE LENGTH

1. Look at **Line A** (separate sheet)
2. Estimate length in cm
3. Repeat for **Line B**

Don't measure!

TASK C: STRING LENGTH

1. Feel **String 1** (loop) without looking
2. Estimate length in cm
3. Repeat for **String 2** (knotted)

Touch only!

TASK D: TEMPERATURE

1. Without checking any devices
2. Estimate room temperature
3. Record in °C

✍ **RECORD YOUR ESTIMATES**

Time:	Line A:	Line B:	String 1:	String 2:	Temp:
_____ sec	_____ cm	_____ cm	_____ cm	_____ cm	_____ °C

Your brain's internal clock

Explore how accurately you perceive time and physical dimensions.

Part A: 60-Second Estimation

1. Close your eyes
2. When you hear “Start”, sense time passing
3. Say “Stop” when you think **60 seconds** have passed
4. Record the actual time

Part B: Visual Estimation (Lines)

1. Look at **Line Card A** — estimate its length in cm
2. Repeat with **Line Card B**

Part C: Tactile Estimation (Strings)

1. Feel **String 1** with eyes closed — estimate its length in cm
2. Repeat with **String 2**

What You’ll Record

- Your 60-second estimate (actual seconds)
- Line estimates (cm)
- String estimates (cm)
- Whether you counted

Task 3: Sixth Sense

DATALAB

👁 SIXTH SENSE

Can you feel when someone is watching?

INSTRUCTIONS

1. **TARGET** sits with back to partner, eyes closed
2. **STARER** flips a coin: **Heads** = stare at back of partner's head; **Tails** = look away
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Confidence: 1 = pure guess, 5 = absolutely certain

✍ **TOTAL CORRECT:** _____ / 6

Can you tell when someone is watching you?

What You'll Do

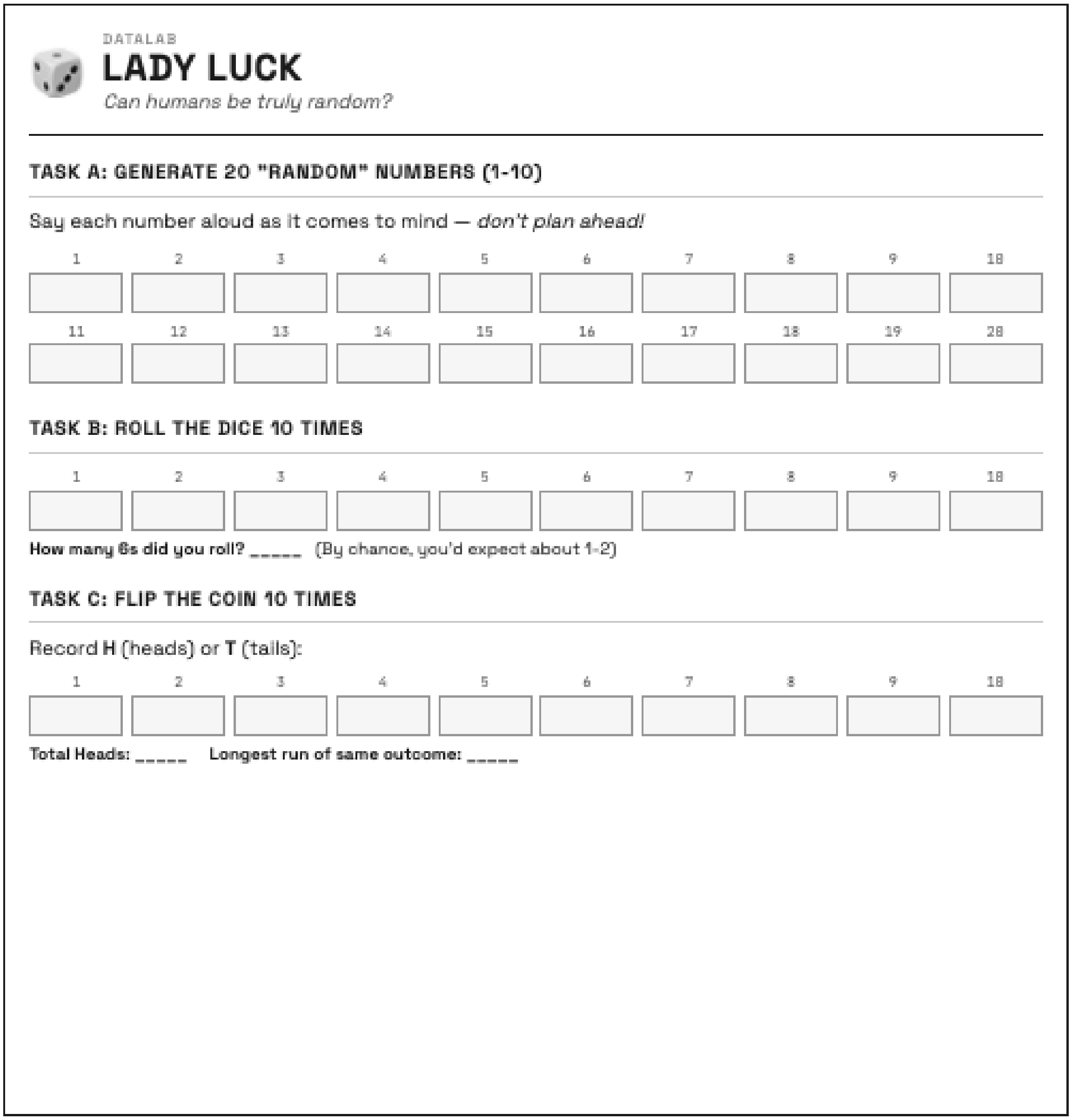
1. Sit **back-to-back** with your partner (~1m apart)
2. **STARER**: Flip the coin
   - Heads = Stare at partner's head
   - Tails = Look away
3. Hold for **15 seconds**
4. **TARGET**: Say "Staring" or "Not staring"

5. Rate your **confidence** (1 = guess, 5 = certain)
6. Complete **6 trials** as target, then swap

What You'll Record

- For each trial: actual, guess, confidence
- Total correct out of 6

Task 4: Lady Luck



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Say each number aloud as it comes to mind — *don't plan ahead!*

1	2	3	4	5	6	7	8	9	10

11	12	13	14	15	16	17	18	19	20

TASK B: ROLL THE DICE 10 TIMES

1	2	3	4	5	6	7	8	9	10

How many 6s did you roll? _____ (By chance, you'd expect about 1-2)

TASK C: FLIP THE COIN 10 TIMES

Record **H** (heads) or **T** (tails):

1	2	3	4	5	6	7	8	9	10

Total Heads: _____ **Longest run of same outcome: _____**

How random can you be?

Part A: Random Numbers

Say **20 “random” numbers** between 1 and 10 aloud. Partner records them.

Part B: Dice Rolling

1. Predict: more 6s than expected by chance?
2. Roll the die **10 times**, record each

Part C: Coin Flipping

1. Flip **10 times**, record each (H or T)
2. Note the longest run

What You’ll Record

- All 20 numbers
- Dice prediction and outcomes
- Coin outcomes and longest run

Task 5: Frontal Lob

DATALAB

FRONTAL LOB

Precision meets confidence

INSTRUCTIONS

1. **Before throwing:** Rate your ball-throwing ability (1 = terrible, 10 = excellent)
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6. *Optional:* Try 5 throws with your non-dominant hand

✍ **RECORD YOUR DATA**

Self-rating:	Dominant hand hits:	Non-dominant hits:	Which hand is dominant?
_____ / 10	_____ / 10	_____ / 5	☐ Left ☐ Right

Precision meets confidence

What You'll Do

1. **Before throwing**: Rate your throwing ability (1–10)
2. Stand at the marked line
3. Throw **10 ping pong balls** at the basket
4. Count your successes

What You'll Record

- Your self-rating (1–10)
- Number of successes (out of 10)
- Which is your dominant hand

After the Tasks

Complete Part 2 Survey

Once all tasks are complete, return to the survey:

- How you're feeling **now** (compare to before)
- Favourite / least favourite task
- Any comments

> ! Pre-Post Design
>
> By measuring your mood before AND after, we can see if DataLab affected how you feel. This is a **within-subjects** design.

Key Takeaways

> i What We Did Today
>
> 1. Completed the memory test together
> 2. Worked through 5 tasks measuring different constructs
> 3. Created a pre-post design to measure mood change
> 4. Generated data we'll analyse throughout the term

Next Week

Week 12: Your first lecture!

- Wednesday 10:00–12:00
- We'll analyse the DataLab results together
- Introduction to distributions and variability

See Week 12 for details.